

CSE422: Artificial Intelligence

Project Name:

Career Switch Prediction

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Group 17

Section: 18

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Table of Contents:

1. Introduction 2
2. Dataset description 3
3. Data pre-processing 4
4. Data splitting 5
5. Model Training and Evaluation 5
6. Model Performance Comparison 5
7. Conclusion 10

Introduction

In a rapidly changing job market, professionals often contemplate switching careers for various reasons such as better opportunities, job dissatisfaction, or skill mismatches. Organizations, on the other hand, face significant challenges in retaining talent and planning workforce development. Predicting potential career switchers can help organizations take proactive measures and guide individuals toward better-suited roles or skill development programs. This project is motivated by the need to understand and predict such career-switching tendencies using machine learning.

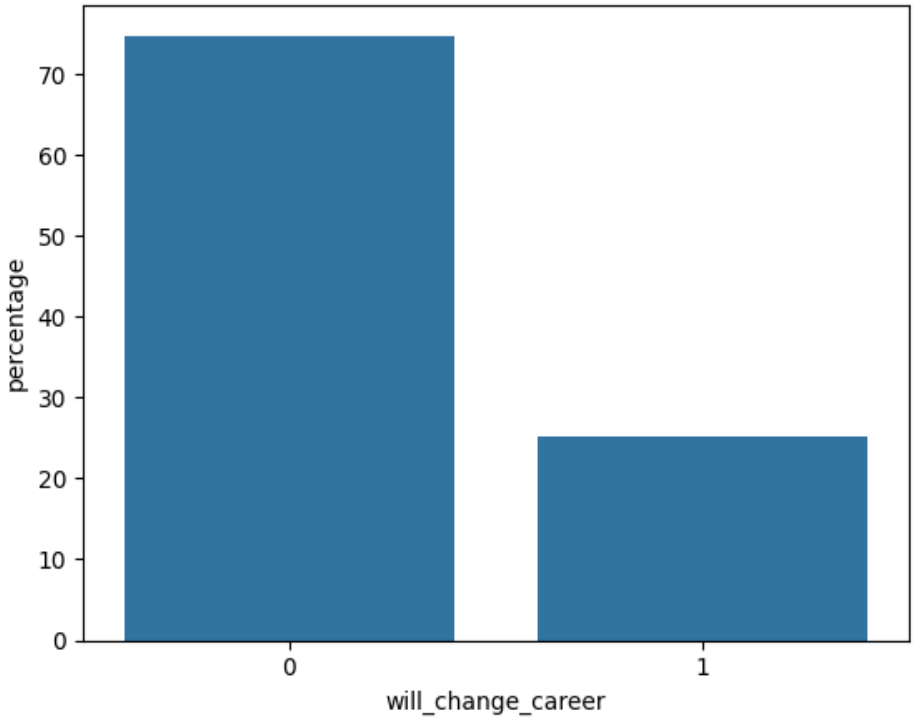
The **Career Switch Classification** project focuses on predicting whether an individual is likely to switch careers based on demographic, educational, and professional attributes. By leveraging data-driven insights, the model aims to assist employers, recruiters, and policymakers in identifying career-change patterns. This can enable better targeting of training programs, improve employee engagement strategies, and enhance the overall understanding of workforce mobility trends.

Dataset Description

The dataset contains 5000 samples and 14 original features, a mix of categorical and numerical variables. The target is binary (0 = no switch, 1 = will switch), making this a classification problem.

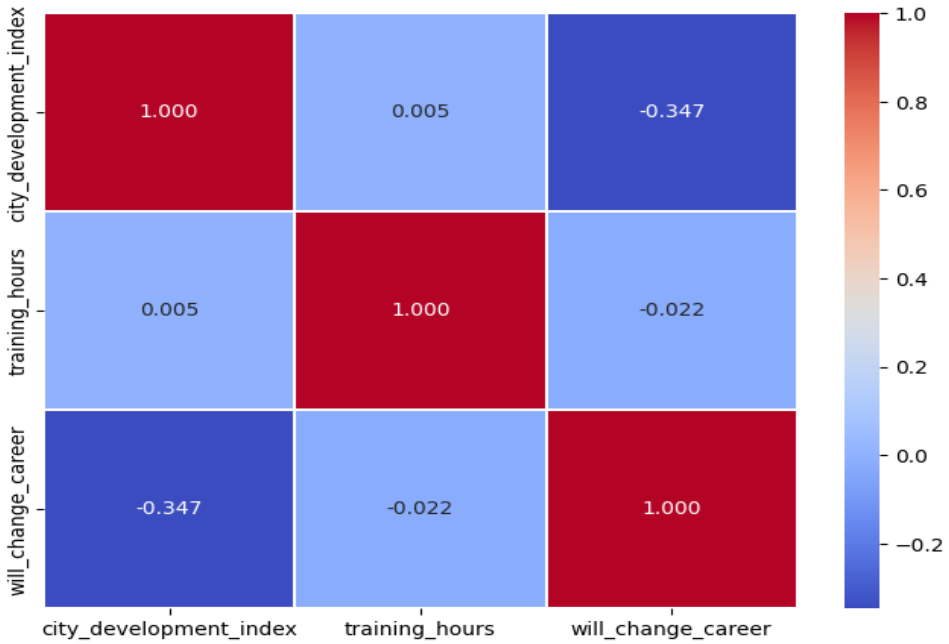
Target Class Distribution

The dataset shows a class imbalance with a majority of individuals not switching careers.



Feature Correlation Heatmap

Correlation between numerical features and target shows some meaningful relationships.



Dataset Preprocessing

The dataset contained several missing values primarily in categorical columns such as gender, education_level, major_discipline, company_size, and company_type. Missing values were filled using a placeholder value -1. For the 'experience' column, string ranges (e.g., '>20', '<1') were mapped to numerical values, and missing entries were imputed with the median value. The values in company size had a typo of “Oct-49” which should’ve been actually “10-49”. We have also replaced the values of the categorical column, “relevant_experience” with a binary encoding, with “Has relevant experience” with 1 and “No relevant experience” with 0.

Categorical columns were encoded using one-hot encoding, increasing the total number of features to 155. Numerical features were standardized using StandardScaler to ensure uniform contribution to distance-based models and stable neural network training.

The enrollee_id column was dropped as it had no predictive value.

Dataset Splitting:

The data was split into training (70%) and test (30%) sets using stratified sampling to maintain the original class distribution in both sets. All preprocessing transformations were applied using parameters derived from the training set to prevent data leakage.

Model Training and Evaluation:

Four models were trained and evaluated: Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and a Neural Network. Performance was measured using accuracy, precision, recall, and ROC AUC scores

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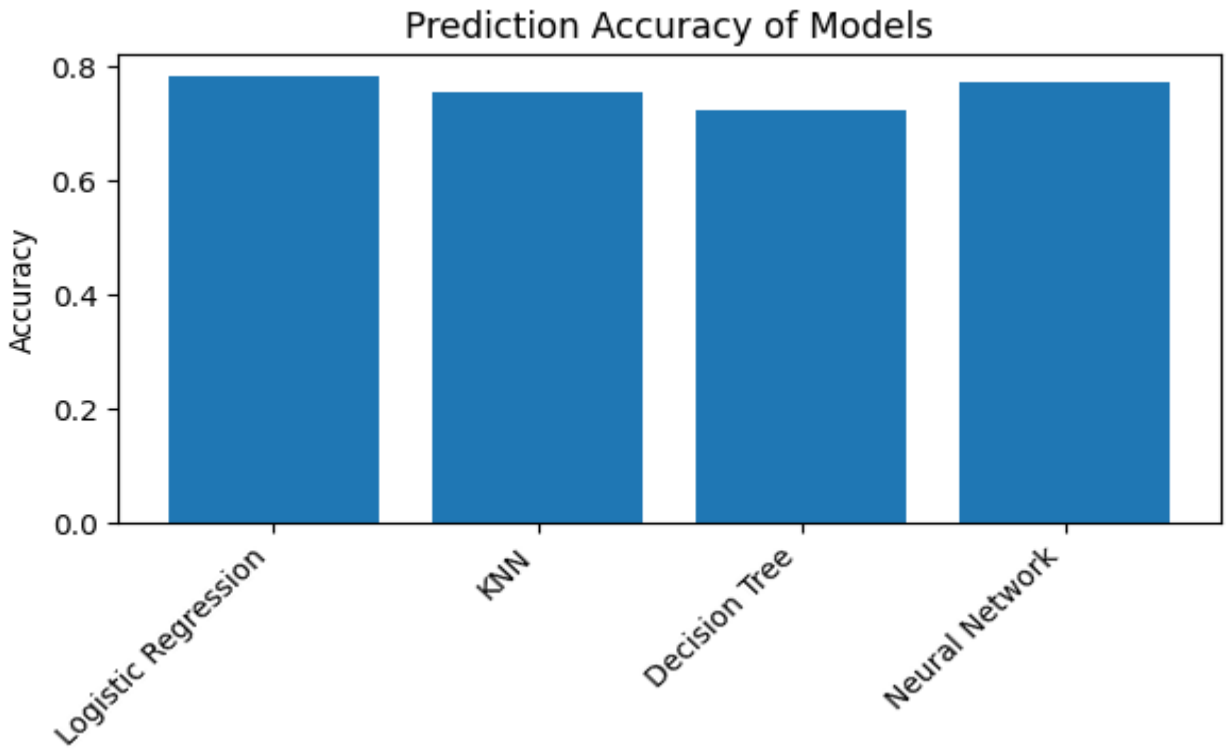
Model Performance Comparison:

The table below compares model performance on the test set:

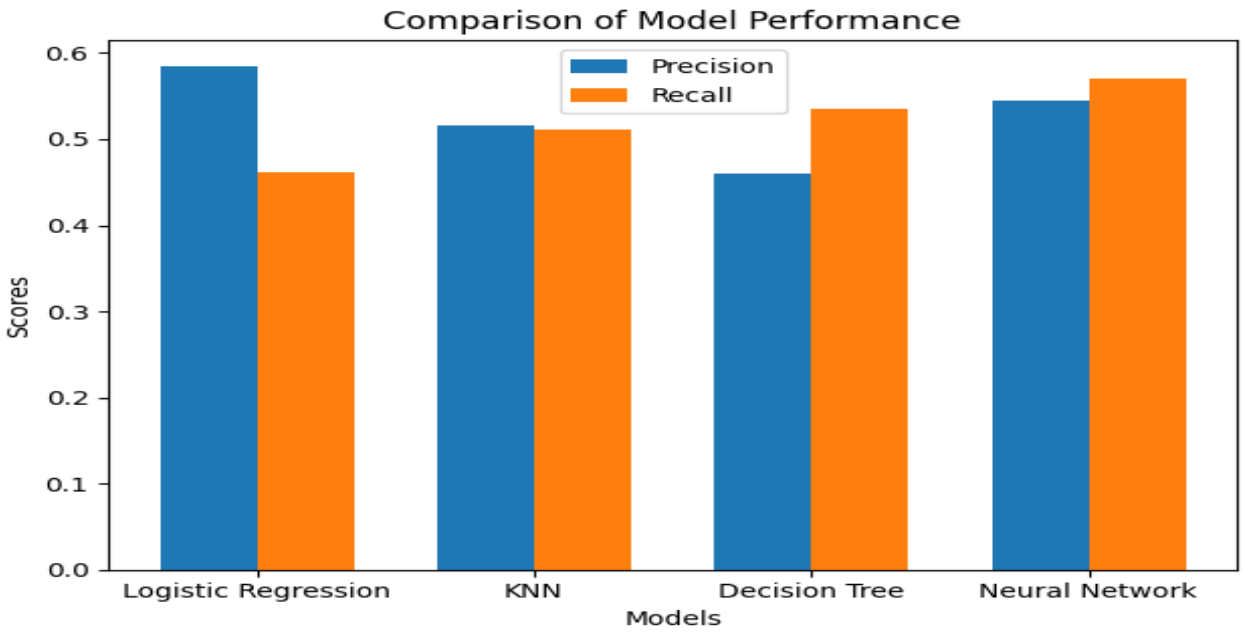
Model	Accuracy	Precision	Recall	ROC/AUC
Logistic Regression	78%	59%	46%	0.76

KNN (k=5)	76%	53%	50%	0.74
Decision Tree	72%	45%	53%	0.65
Neural Network	77%	50%	48%	0.75

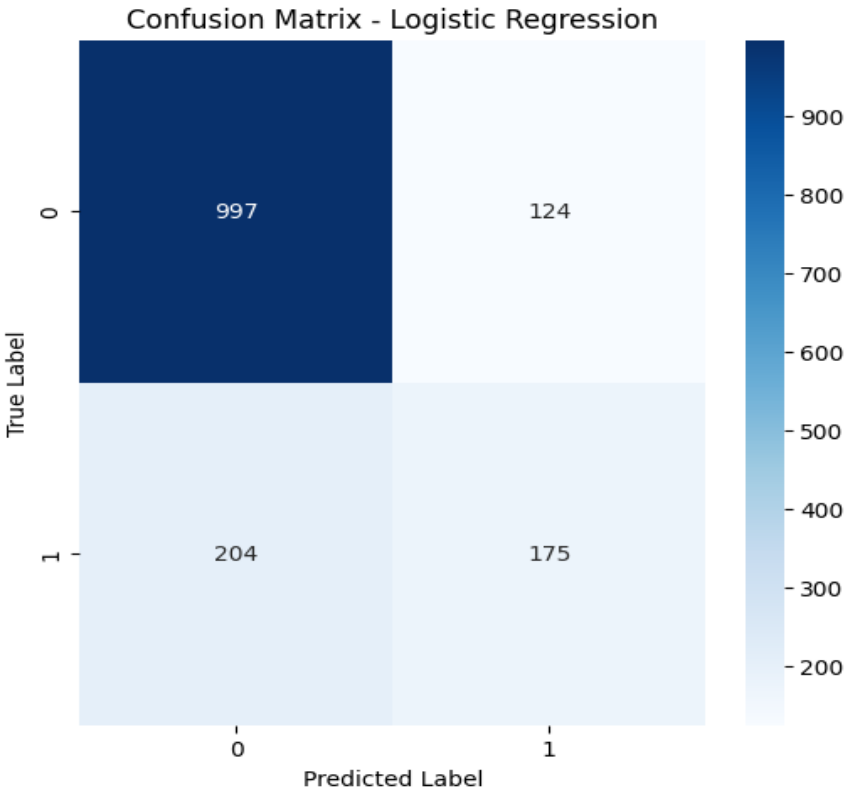
Prediction Accuracy Of Models:

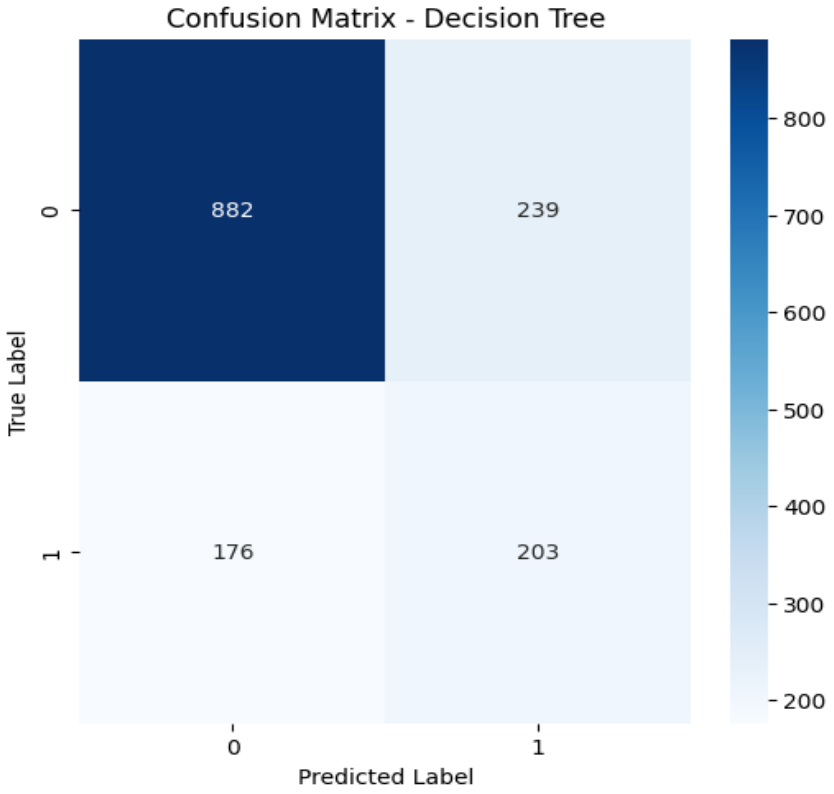


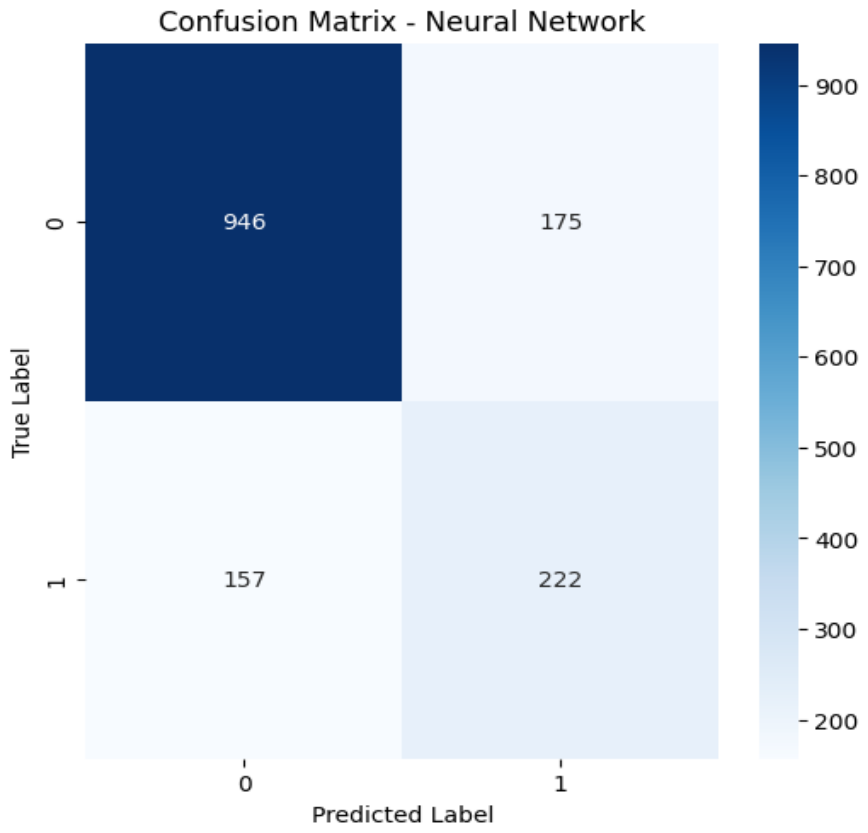
Comparison Of Model Performance:

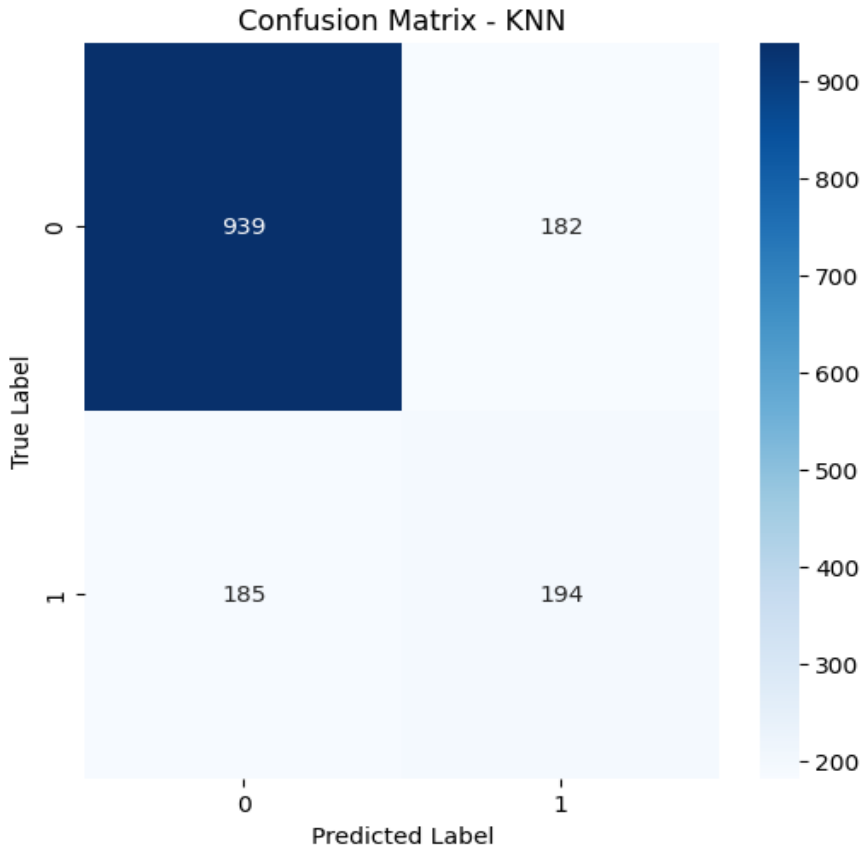


Confusion Matrix: Logistic Regression:

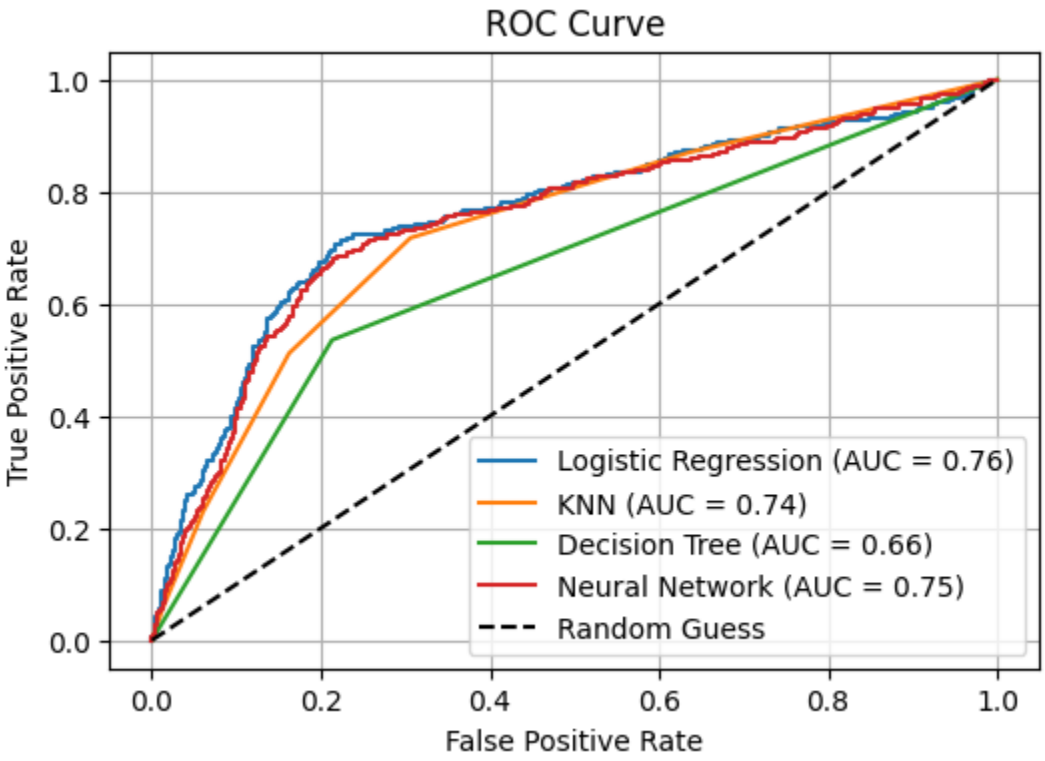





Confusion Matrix - Neural Network
946
175
157
222
0
1
True Label
0
1
Predicted Label
900
800
700
600
500
400
300
200


Confusion Matrix - KNN
939
182
185
194
0
1
True Label
0
1
Predicted Label
900
800
700
600
500
400
300
200

ROC Curve Comparison:



Conclusion:

The project demonstrated the feasibility of predicting career-switch tendencies using basic demographic and professional information. Logistic Regression achieved the highest overall performance in terms of accuracy and AUC. KNN and the neural network showed better recall, while the Decision Tree provided insights but suffered from overfitting.

Key challenges included handling imbalanced data, transforming categorical variables, and ensuring model generalization. Future enhancements could include hyperparameter tuning, ensemble methods, and addressing class imbalance using SMOTE or cost-sensitive learning.